

Sachin Reddy Kunta

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EDUCATION

New York University

New York, NY

Master of Science in Computer Engineering

Sep 2023 – May 2025

- **Relevant Coursework:** Machine Learning, Deep Learning, Databases, Computer Systems Architecture, Internet Architecture and Protocols, Network Security, Real-Time Embedded Systems

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C++

Frameworks/Libraries: Node.js, Express, React, Next.js, Spring Boot

Cloud/Infrastructure: AWS, GCP, Azure, Docker, Kubernetes, Terraform, GitHub Actions

Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB

Other Tools: Kafka, GraphQL, gRPC, Prometheus, Grafana, Nginx

EXPERIENCE

Mobile Premier League

Bangalore, India

Software Development Engineer

Jun 2021 – Jul 2023

- Developed and maintained backend APIs (Java, Node.js, gRPC) supporting 10M+ daily users for payments and fraud detection, leveraging PostgreSQL, Redis, and containerized microservices to ensure scalability.
- Created full-stack internal dashboards (React + Node.js) for operations teams to manage refunds, user verifications, and rewards — saving 20+ weekly engineering hours and enabling faster resolutions.
- Led architecture migration of legacy internal tools to modular micro-frontend and backend systems; trained engineering teams and improved deployment velocity by 70%.
- Integrated secure payment gateways (Stripe, Razorpay, UPI) to enable seamless checkout, increasing success rates and improving user transaction reliability.
- Enhanced developer velocity by implementing containerized local environments (Docker Compose) and observability stack (Grafana, AWS CloudWatch), cutting onboarding time by 50% and improving incident debugging.

Masai School

Bangalore, India

Software Development Intern

Dec 2020 – May 2021

- Developed an objective assessment platform using React.js, Node.js, and PostgreSQL, integrating Monaco Editor for real-time coding execution and optimized RESTful API performance.
- Built scalable leaderboard and contest management system to handle high-volume code submissions, using Redis caching for efficient ranking and result evaluation across thousands of users.
- Implemented secure authentication using JWT and bcrypt, and enforced role-based access control aligned with OAuth 2.0 principles to safeguard user sessions and admin operations.

PROJECTS

Supervised Fine-Tuning of LLaMA 3-8B Model | *PyTorch, LoRA, Google Colab*

Sep 2024 – Dec 2024

- Fine-tuned LLaMA 3-8B model to predict math question correctness using Supervised Fine-Tuning.
- Implemented LoRA and quantization techniques for efficient training, achieving a test accuracy of 84.43%.
- Applied prompt engineering and hyperparameter tuning to optimize model performance.
- Conducted training on Google Colab with the Unsloth framework for resource-efficient experimentation.

Video Captioning with MSR-VTT Dataset | *PyTorch, MSR-VTT, BART, T5, CLIP*

Sep 2024 – Dec 2024

- Developed a video captioning model using the MSR-VTT dataset, integrating visual features with generative models like BART and T5.
- Utilized CLIP-based embeddings for feature extraction and implemented attention mechanisms to enhance caption relevance and achieved significant improvements in caption accuracy, evaluated using BLEU and METEOR scores.

Real-Time Collaborative Text Editor | *Node.js, Express, React, Socket.io*

Jan 2024 – May 2024

- Designed and implemented a real-time collaborative text editor using Node.js, Socket.io and React, enabling multiple users to edit documents simultaneously.
- Developed conflict resolution strategies using Operational Transforms to handle concurrent edits and implemented user authentication and permission controls for secure document access.